Reinforcement Learning

Agents that learn by interacting with environments - from bandits to deep RL and RLHF.

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1. RL basics: agent, environment, state, action, reward, policy [Beginner]

Explanation (English)

An agent observes a state, takes an action, gets a reward and next state; goal is to maximise cumulative discounted return.

Tanglish (Tamil + English) Vilakkam

Agent action edukkum, environment reward tharum. Total reward maximize.

Example / Program

```
import gymnasium as gym
e=gym.make('CartPole-v1');s,_=e.reset(seed=0);tot=0
for _ in range(200):
    s,r,t,tr,_=e.step(e.action_space.sample());tot+=r
    if t or tr: break
print(tot)
```

Learn more

YouTube search: RL basics: agent, environment, state, action, reward, policy tutorial

2. Multi-armed bandits, exploration vs exploitation [Beginner]

Explanation (English)

Epsilon-greedy, UCB, Thompson sampling balance trying new options and using the best known.

Tanglish (Tamil + English) Vilakkam

Explore vs exploit. epsilon-greedy.

Example / Program

```
import numpy as np
q=np.zeros(3);n=np.zeros(3);p=[.2,.5,.8]
for _ in range(2000):
    a=np.random.randint(3) if np.random.rand()<.1 else q.argmax()
    r=float(np.random.rand()<p[a]);n[a]+=1;q[a]+=(r-q[a])/n[a]
print(q.round(2))
```

Learn more

YouTube search: Multi-armed bandits, exploration vs exploitation tutorial

3. Markov Decision Processes, Bellman equations [Intermediate]

Explanation (English)

MDP = (S, A, P, R, gamma). V(s)=max_a [R + gamma * sum P V(s')] - the Bellman optimality equation.

Tanglish (Tamil + English) Vilakkam

MDP, Bellman equation - RL-oda maths base.

Learn more

YouTube search: markov decision process bellman equation explained

4. Dynamic programming: policy and value iteration [Intermediate]

Explanation (English)

Solve known MDPs by iterating policy evaluation/improvement or value updates.

Tanglish (Tamil + English) Vilakkam

Policy iteration, value iteration.

Example / Program

```
import numpy as np
V=np.zeros(2);P=np.array([[.9,.1],[.2,.8]]);R=np.array([1,0])
for _ in range(200): V=R+.9*P@V
print(V)
```

Learn more

YouTube search: Dynamic programming: policy and value iteration tutorial

5. Monte Carlo and Temporal-Difference learning, SARSA, Q-learning [Intermediate]

Explanation (English)

Learn from experience without a model. Q-learning: Q(s,a) += alpha*(r + gamma*max Q(s',a') - Q(s,a)).

Tanglish (Tamil + English) Vilakkam

Q-learning table-la values update pannum.

Example / Program

```
import gymnasium as gym,numpy as np
e=gym.make('FrozenLake-v1',is_slippery=False);Q=np.zeros((16,4))
for ep in range(2000):
    s,_=e.reset()
    for _ in range(100):
        a=np.random.randint(4) if np.random.rand()<.2 else Q[s].argmax()
        s2,r,t,tr,_=e.step(a);Q[s,a]+=.1*(r+.95*Q[s2].max()-Q[s,a]);s=s2
        if t or tr: break
print(Q.argmax(1))
```

Learn more

YouTube search: Monte Carlo and Temporal-Difference learning, SARSA, Q-learning tutorial

6. Deep Q-Networks (DQN) and variants [Advanced]

Explanation (English)

Neural network approximates Q; replay buffer + target network stabilise; Double/Dueling/Prioritised/Rainbow variants.

Tanglish (Tamil + English) Vilakkam

DQN: Q-table-ku bathil neural network.

Example / Program

```
from stable_baselines3 import DQN
# DQN('MlpPolicy','CartPole-v1').learn(20000)
```

Learn more

YouTube search: Deep Q-Networks (DQN) and variants tutorial

7. Policy gradients: REINFORCE, Actor-Critic, A2C/A3C [Advanced]

Explanation (English)

Directly optimise the policy with gradient ascent on expected return; critics reduce variance.

Tanglish (Tamil + English) Vilakkam

Policy directly optimize - REINFORCE, actor-critic.

Learn more

YouTube search: policy gradient reinforce actor critic explained

8. PPO, TRPO, SAC, DDPG, TD3 [Advanced]

Explanation (English)

Modern algorithms. PPO is the practical default; SAC/TD3 for continuous control.

Tanglish (Tamil + English) Vilakkam

PPO most popular. SAC/TD3 continuous actions.

Example / Program

```
from stable_baselines3 import PPO
m=PPO('MlpPolicy','CartPole-v1',verbose=0).learn(10000)
```

Learn more

YouTube search: PPO, TRPO, SAC, DDPG, TD3 tutorial

9. Model-based RL, planning, MCTS, AlphaGo/AlphaZero/MuZero [Advanced]

Explanation (English)
Learn or use a model of the environment to plan; MCTS + neural networks mastered Go and chess.

Tanglish (Tamil + English) Vilakkam
AlphaGo - MCTS + neural net.

Learn more
YouTube search: alphazero explained mcts

10. Multi-agent RL, imitation learning, offline RL, inverse RL [Advanced]

Explanation (English)
Cooperation/competition among agents, learning from demonstrations, learning from fixed datasets.

Tanglish (Tamil + English) Vilakkam
Multi-agent, imitation, offline RL.

Learn more
YouTube search: multi agent reinforcement learning introduction

11. RLHF and RL for LLM reasoning (PPO, GRPO, DPO) [Advanced]

Explanation (English)
Use human or verifiable rewards to align and improve reasoning in language models.

Tanglish (Tamil + English) Vilakkam
RLHF, GRPO - LLM-a improve panna RL.

Learn more
YouTube search: grpo rlhf reasoning models explained

12. RL projects: CartPole, FrozenLake, Atari, trading bot, robot sim [Intermediate]

Explanation (English)
Start with Gymnasium classic control, then Atari/MuJoCo; always plot learning curves over multiple seeds.

Tanglish (Tamil + English) Vilakkam
CartPole -> Atari -> custom env.

Learn more
YouTube search: reinforcement learning project gymnasium stable baselines3